

HURDLE 5

Deploy to Azure & Shift-Left Green Score

Deploy your API to Azure App Service and implement the sustainability gate.

 30 Minutes

Objective

Deploy the FastAPI application to Azure App Service so it is accessible via a public URL. Then implement the GreenOps Shift-Left Green Score — the A–F sustainability rating that simulates a CI/CD pipeline gate based on predicted emissions.

Background

All the prior hurdles built towards this moment. Deployment closes the loop: your sustainability data is now live on the cloud. The Green Score completes the 'Shift-Left' concept — rather than discovering a project's carbon impact after deployment, the score flags it automatically during the build, giving developers a chance to optimise before it is too late.

Tasks

Task A — Prepare for Deployment

1. Ensure requirements.txt is up to date (pip freeze > requirements.txt).
2. Create a startup command file: Procfile (for App Service) with content: web: uvicorn api.main:app -host 0.0.0.0 --port 8000
3. Set DATASET_PATH and MODEL_PATH as environment variables in your code instead of hardcoded paths.
4. Commit all changes and push to GitHub.

Task B — Deploy to Azure App Service

5. In the Azure Portal → App Services → + Create.
6. Select resource group: greenops-rg-[roll-number].
7. Runtime: Python 3.11 | OS: Linux | Region: East US | Plan: Free (F1).
8. Deployment method: GitHub Actions (connect your repo) OR use Azure CLI:

```
az webapp up --name greenops-api-[rollnumber] \ --resource-group
greenops-rg-[rollnumber] \ --runtime 'PYTHON:3.11'
```

- Wait for deployment to complete. Visit [https://greenops-api-\[rollnumber\].azurewebsites.net/health](https://greenops-api-[rollnumber].azurewebsites.net/health)
- Confirm it returns: {"status": "ok"}
- Screenshot the live URL response.

Task C — Implement the Green Score

Add a new endpoint GET /green-score to your FastAPI app. Use the scoring table below:

Grade	Threshold	Recommended Action	Pipeline Gate
A	< 2 kg CO2e/day average	Excellent — no action needed	● PASS
B	2–5 kg CO2e/day average	Good — minor optimisation advised	● PASS
C	5–10 kg CO2e/day average	Moderate — review VM sizing	● PASS
D	10–20 kg CO2e/day average	Poor — immediate rightsizing required	● WARNING
F	> 20 kg CO2e/day average	Critical — pipeline soft gate triggered	● BLOCKED

- The endpoint should calculate the average daily CO2e from the dataset, determine the grade, and return: {"grade": "B", "avg_daily_co2e": 3.7, "action": "...", "gate": "PASS"}
- Update the Streamlit dashboard to call this endpoint and display the Green Score badge prominently.
- Redeploy after adding the new endpoint.

Task D — Final Commit & Documentation

- Update README.md with: project description, architecture diagram (ASCII is fine), setup instructions, live API URL, and your Green Score result.
- Final commit: 'feat: Azure deployment + Green Score shift-left gate'

Concept Check

- What is the 'Shift-Left' principle in DevOps, and how does the Green Score apply it to sustainability?
- What is Azure App Service and how is it different from running on a VM (IaaS)?
- If a project scores F, what specific infrastructure changes would you recommend first?

Final Self-Evaluation Checklist

☐	FastAPI app deployed to Azure App Service and /health returns 200 OK
☐	Live public URL accessible and tested from browser
☐	GET /green-score endpoint implemented and returns correct grade for the dataset
☐	Green Score badge visible on Streamlit dashboard
☐	README updated with architecture, setup instructions, and live URL
☐	All code committed and pushed to GitHub
☐	Concept check answered

 **SPRINT COMPLETE** — Well done! Prepare a 2-minute demo for the retrospective.

Retrospective Prompts

Be ready to answer any one of these during the group wrap-up:

- Walk us through your deployed URL and one metric you're proud of.
- What was the hardest hurdle and what did you do to unblock yourself?
- What would you add to the GreenOps Dashboard if you had one more week?
- Where in a real DevOps pipeline would you integrate the Green Score gate?